MLFF-DATA2A Automatic Review-Manifest Inference Gate

Implemented in mdstats 0.20.60a0

Purpose

The campaign manifest is the reviewed boundary between raw file discovery and scientific data preparation. MLFF-DATA2A removes repetitive manual annotation without weakening that review boundary. It separates three classes of evidence:

1. **Observed metadata** recovered from `vasprun.xml` controls and cells.
2. **Filename candidates** that describe intended strain semantics.
3. **Promoted operational assertions** allowed to influence reference-cell resolution, partitioning, condition coverage, and reporting only after geometry verification succeeds.

A filename is never sufficient proof of strain. Manual approval remains mandatory after inference.

Public records

- `ManifestInferencePolicy`
- `ManifestInferenceResult`
- `TrainingDataRunSpec.inference`
- `infer_training_manifest_metadata(...)`

The immutable policy records tolerances and filename interpretation. The result records counts, warnings, the inferred manifest digest, and the policy digest.

XML-derived metadata

One bounded-memory tolerant XML pass recovers completed records for:

- `TEBEG`, `TEEND`, target temperature;
- `POTIM`, `NSW`;
- `IBRION`, `MDALGO`, `SMASS`, `ISIF`, `LANGEVIN_GAMMA`;
- NVT/NVE ensemble and thermostat classification when controls are resolvable;
- initial and per-calculation cell matrices;
- ordered atom symbols;
- fixed-cell status and maximum relative cell deviation.

A truncated XML may populate the **review manifest** from completed records and must carry an explicit parse warning. This tolerant pass does not replace the later strict DATA2 source-quality, label, and trajectory qualification gate.

Filename strain grammar

The LTA profile recognizes filenames containing:

```
_strained.hydro+VALUE
_strained.ortho-VALUE
_strained.shear+VALUE
```

An explicit `%` is a percentage. By default, an unmarked absolute value at or above one is also interpreted as a percentage:

```
hydro+5    -> +0.05 relative volume change
ortho-2    -> -0.02 signed axial delta
shear+2    -> +0.02 engineering shear gamma
```

Values below one retain fractional meaning:

```
hydro+0.05 -> +0.05 relative volume change
```

Temperature tokens may be present or absent in either the strained filename or its reference filename. They are excluded from identity matching and used only as a ranking hint after geometry passes.

Fixed-cell prerequisite

Static strain inference is allowed only when both the strained trajectory and candidate reference are verified fixed-cell MD. The maximum trajectory cell deviation must satisfy

$$\max_t \frac{\|H_t - H_0\|_F}{\|H_0\|_F} \leq \varepsilon_{\text{fixed}}.$$

Variable-cell runs remain ungrouped.

Exact LTA strain definitions

The definitions are frozen to the supplied six-strain generator.

Hydrostatic

A filename value $\Delta V/V$ denotes a volume ratio

$$\det F = 1 + \Delta V/V,$$

with equal axial stretch

$$U_{\text{expected}} = (1 + \Delta V/V)^{1/3} I.$$

Orthorhombic

For signed d, the deformation in LTA conventional axes is

$$U_{\text{aligned}} = \operatorname{diag}\left(1+d, 1-d, \frac{1}{1-d^2}\right),$$

which preserves volume exactly.

Shear

For signed engineering shear γ, form the simple-shear matrix

$$S = I + \gamma\, e_x \otimes e_y.$$

The applied deformation is its exact symmetric right-polar stretch

$$U_{\text{aligned}} = (S^T S)^{1/2}.$$

The LTA primitive-cell conventional axes are reconstructed from primitive rows a, b, c as $b + c - a$, $a + c - b$, and $a + b - c$, normalized and verified orthogonal before transforming the expected stretch back to Cartesian space.

Reference resolution

A candidate reference must be unstrained and satisfy:

- filename identity after removing the strain tag and temperature tokens;
- identical ordered atom identities;
- verified fixed-cell status;
- a cell deformation that passes the exact expected strain test.

Among geometry-passing candidates, target-temperature distance is a ranking hint. A unique closest candidate is selected. Equally ranked equivalent cells may form a consensus. Equally ranked non-equivalent cells are ambiguous and remain unpromoted.

Verification and fail-safe behavior

For reference cell H_0 and strained cell H_s, mdstats calculates

$$F = H_s H_0^{-1} = RU$$

and compares the observed right stretch U, volume ratio, and rotation with the expected profile deformation.

Only a passing relationship receives:

- `reference_group`;
- optional `reference_run_id`;
- `intended_strain_class` and magnitude/sign assertions;
- a verified reference-cell assertion on the selected baseline.

If verification fails or is ambiguous:

- no operational strain assertion is promoted;
- `reference_group` and `reference_run_id` remain null for that relationship;
- candidate matrices, residuals, references, and rejection reasons remain in `inference` for user inspection;
- `prepare` emits a warning and still requires manifest approval.

A failed sibling does not erase an independently verified relationship.

CLI contract

The first `prepare` invocation discovers sources, runs inference, writes the populated manifest, prints a compact resolution summary, and stops for review.

```
python tools/mdstats-mlff-campaign.py --config campaign.toml prepare
python tools/mdstats-mlff-campaign.py --config campaign.toml prepare --approve-manifest
```

Use `--refresh-inferences` before approval after source files or inference policy settings change.

Production acceptance

A production archive passes this inference gate when:

- every intended fixed-cell strain has one verified relationship;
- no intended strain is rejected or ambiguous;
- XML parse warnings are retained for later source-quality review;
- the user approves the exact populated manifest digest.